

Intelligent Traffic Risk Prediction

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Abstract

Road traffic crashes are a leading cause of death and injury worldwide, and heterogeneous, weakly-lane-disciplined urban traffic—where pedestrians, cyclists, motorcycles, auto-rickshaws, cars, buses, trucks, and light commercial vehicles interact simultaneously—is especially difficult to assess proactively. Classical surrogate safety measures such as Time-to-Collision (TTC) and Post-Encroachment Time (PET) evaluate road users mostly in isolated pairs, assume constant velocity, and reduce risk to a threshold-based safe/unsafe decision, which loses information about how risk evolves and struggles under the negotiation dynamics typical of mixed traffic. This study asks whether a graph-based, physics-guided, continuous-risk framework can extend recent scene-level pedestrian–vehicle conflict prediction work (Muduli et al., 2026) to the broader eight-class heterogeneous traffic environment recorded at a real Indian intersection. Roadside CCTV video of the Main Building four-road junction at the Indian Institute of Technology Kharagpur was processed through a perception pipeline (YOLOv8 detection, ByteTrack multi-object tracking, ROI filtering, homography-based Bird’s-Eye-View transformation) to obtain metric trajectories for eight road-user classes. Trajectories were smoothed with a physics-informed neural network and converted into per-frame dynamic interaction graphs (15 m interaction radius). A braking-aware Modified Time-to-Collision, combining conventional TTC with a class-dependent braking-feasibility ratio, was used to construct a continuous future-risk regression target via an exponential decay mapping. A hybrid Graph Transformer and Temporal Convolutional Network (ST-GNN) was trained to jointly predict continuous pairwise interaction risk and scene-level risk over a chronologically split train/validation/test partition (4,197 / 899 / 900 graph frames). A locally deployed LLM (Llama 3 via Ollama) converts model outputs into structured JSON safety reports only when a scene-risk alert threshold is crossed; it plays no role in prediction. On the held-out chronological test set, the regression model achieved Pairwise MAE = 0.112, Pairwise R^2 = 0.558, Pairwise RMSE = 0.184, Pairwise MSE = 0.034; Scene-level MAE = 0.105, Scene-level R^2 = 0.043, Scene-level RMSE = 0.140, Scene-level MSE = 0.020. A focused ablation isolating the braking-aware Modified TTC against a standard constant-velocity TTC baseline showed a reduction in MAE from approximately 0.22 to 0.11, indicating that braking-feasibility information materially improves the surrogate risk target. Secondary threshold-based discrimination analysis (a downstream, non-primary evaluation) gave ROC-AUC = 0.909, Recall = 83.79% at τ = 0.3, and Specificity = 98.5% at τ = 0.7. Measured inference latency was 15.48 ms/frame (64.6 FPS) on an NVIDIA RTX 3080 GPU. The continuous, physics-guided risk formulation and the graph-based heterogeneous representation together provide a workable extension of scene-level pedestrian–vehicle conflict modelling to full mixed-traffic environments, while the LLM layer converts numerical output into operator-usable text without participating in the underlying prediction. The moderate scene-level R^2 (0.043) indicates that scene-level aggregation is harder to fit than pairwise risk and is flagged as a limitation requiring further investigation rather than presented as a strength.

Keywords: Traffic Safety Analytics; Graph Neural Networks; Spatio-Temporal Graph Learning; Multi-Agent Interaction Modeling; Time-to-Collision; Physics-Informed Learning; Explain-

able AI; Large Language Models; Heterogeneous Urban Traffic; Bird's-Eye-View Trajectory Estimation.

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1 Introduction

Road traffic injuries remain one of the most persistent public-health and transportation problems worldwide. The World Health Organization estimates roughly 1.19 million deaths and 20–50 million non-fatal injuries annually from road crashes (WHO, 2023), and India alone recorded close to 168,500 traffic fatalities and over 460,000 accidents in 2022 (MoRTH, 2022). Vulnerable road users—pedestrians, cyclists, and motorcyclists—bear a disproportionate share of this burden because they lack the structural protection of an enclosed vehicle.

Traditional safety analysis is retrospective: it studies historical crash records, which are under-reported, delayed, and inconsistent. This has motivated a shift toward proactive safety assessment, in which surrogate indicators are used to flag hazardous interactions before they become crashes. The two most established surrogate measures are Time-to-Collision (Hayward, 1972) and Post-Encroachment Time (Cooper, 1984). Both are useful but limited in real-time predictive use: TTC assumes constant velocity and evaluates one pair at a time; PET is retrospective by construction and cannot anticipate a conflict before it develops. In dense, heterogeneous, weakly-lane-disciplined traffic—pedestrians, cyclists, motorcycles, auto-rickshaws, cars, buses, trucks, and light commercial vehicles sharing the same space with limited formal priority rules—these simplifying assumptions break down more severely than in signal-controlled, homogeneous traffic.

Recent graph-based learning work has begun to address part of this gap. Muduli, Ghosh and Ukkusuri (2026) propose a multi-relational spatio-temporal graph neural network for predicting safety-critical pedestrian–vehicle interactions at unsignalized crosswalks, framing the problem as binary conflict classification and reporting strong performance (accuracy 90.6%, F1 0.906, AUC 0.950) on a purpose-built Indian dataset. That work is the direct conceptual ancestor of the present project and is used throughout this manuscript as the structural and methodological reference point, not as a source of numbers or claims for this project’s own results.

The present work extends that idea in three specific ways that are supported by the project’s own materials: (i) it moves from a pedestrian-vehicle-only, binary-classification setting to an eight-class heterogeneous traffic representation; (ii) it replaces binary conflict labels with a continuous, physics-guided risk target built from a braking-aware Modified TTC; and (iii) it adds a downstream, non-predictive LLM layer that converts grounded numerical model output into structured, human-readable safety reports. These three extensions are the paper’s central, defensible contributions; broader claims (e.g., about generalization across sites, weather robustness, or real-time deployment readiness) are explicitly out of scope and are listed under Limitations and Future Work rather than claimed as results.

1.1 Contributions

- A perception-to-report pipeline that converts heterogeneous roadside CCTV video into metric Bird’s-Eye-View trajectories and dynamic interaction graphs for eight road-user classes.
- A braking-aware Modified Time-to-Collision formulation that folds class-dependent stopping capability into the conventional TTC and converts it into a continuous, bounded risk target via exponential decay.
- A dual-output Graph Transformer and Temporal Convolutional Network (ST-GNN) that jointly predicts pairwise interaction risk and scene-level risk from sequences of dynamic graphs.
- An empirical demonstration, on a chronologically held-out test partition, that the braking-

aware risk formulation measurably outperforms a standard constant-velocity TTC baseline (MAE 0.11 vs. 0.22).

- A downstream, evidence-grounded LLM reporting layer that is invoked only when a scene-risk alert threshold is crossed, and that is explicitly documented as playing no role in prediction.

2 Related Work and Research Gap

2.1 Surrogate Safety Measures

Time-to-Collision (TTC) (Hayward, 1972) and Post-Encroachment Time (PET) (Cooper, 1984) remain the dominant surrogate indicators for proactive conflict analysis (Gettman & Head, 2003; Tarko, 2018). Both were designed for retrospective or simplified pairwise analysis and are known to misjudge risk under braking, acceleration, or near-parallel motion (Li et al., 2022; Sheykhfard et al., 2023).

2.2 Graph-Based Conflict Prediction: The IIT Roorkee Reference Framework

Muduli, Ghosh and Ukkusuri (2026) represent an unsignalized-crosswalk scene as a multi-relational spatio-temporal graph with three explicit relation types—Pedestrian–Pedestrian (PP), Vehicle–Vehicle (VV), and Pedestrian–Vehicle (PV)—each carrying distance- and displacement-based adjacency matrices. Their architecture (MRTGCN) combines Disentangled Multi-Scale Aggregation (DMSA), which bins pairwise distances to avoid over-smoothing, with a Temporal Convolutional Network (TPCNN) backbone, and predicts a binary safety-critical/safe-passage label at each timestep. On a 720-minute, three-site Indian dataset with observer-labelled ground truth (Fleiss’ $\kappa = 0.79$ inter-rater agreement), their best configuration (MRTGCN + DMSA + TPCNN) achieves 90.6% accuracy, F1 = 0.906, precision = 0.927, recall = 0.886, specificity = 0.928, and AUC = 0.950, outperforming GRU, LSTM, and Transformer-MLP baselines. Ablation results in that paper confirm that both MRAM and DMSA contribute measurably to performance, and cross-site transfer experiments show that a small amount (30–40%) of target-site data substantially closes the domain-shift gap. This is a strong, directly relevant precedent, but its scope is deliberately narrow: pedestrian–vehicle conflict at unsignalized crosswalks, framed as binary classification, evaluated with classification metrics.

2.3 Research Gap Addressed by This Project

Three gaps in the reference framework and in the broader literature motivate the present project:

1. **Traffic heterogeneity.** Real urban roads in India rarely involve only pedestrians and vehicles approaching a crosswalk; they involve eight or more distinct road-user classes moving simultaneously under weak lane discipline. A framework built specifically around PP/VV/PV relation types does not directly generalize to this broader taxonomy without modification.
2. **Binary vs. continuous risk.** Binary conflict labels discard information about how risk evolves gradually as an interaction develops, and force a hard decision boundary that may not reflect the physically continuous nature of closing distance and braking capability.
3. **From perception to explanation.** Most graph-learning conflict-prediction systems,

including the reference framework, stop at a predictive score. They do not close the loop to an operator-usable explanation of why a given interaction is flagged.

The present project’s contributions (Section 1.1) map directly onto these three gaps: heterogeneous eight-class graph representation, continuous braking-aware risk regression, and a downstream LLM reporting layer.

2.4 Novelty and Contribution Analysis Relative to the Reference Framework

Table 1: Conservative, item-by-item novelty comparison against Muduli et al. (2026). No component is claimed as invented; every row states the specific, defensible extension made in this project.

Aspect	Reference (Muduli et al., 2026)	This project	Framing
Primary setting	Pedestrian–vehicle interactions at unsignalized crosswalks	Heterogeneous multi-agent urban traffic (8 classes)	Extension of scene-level graph reasoning to a broader road-user taxonomy
Relation structure	Explicit PP/VV/PV adjacency matrices (distance + displacement)	Single interaction-radius graph (15 m) with class one-hot node/edge features	Simplified relation structure—a design trade-off, not an unambiguous improvement
Prediction target	Binary safety-critical conflict classification	Continuous pairwise and scene-level risk regression, with optional operational thresholding	Preserves gradual risk evolution before any decision threshold
Spatial model	Multi-Relational Adjacency Matrices + DMSA	Graph Transformer with learned attention over interaction features	Architectural substitution; DMSA-equivalent benefit not separately ablated
Temporal model	Temporal CNN (TPCNN) vs. LSTM/GRU	Temporal Convolutional Network (TCN) on aligned per-track embeddings	Directly comparable design choice, motivated by the reference paper’s own ablation
Physics layer	Not present	Braking-aware Modified TTC folded into an exponential continuous-risk target	Genuine methodological addition
Perception pipeline	Pre-extracted trajectories (DataFromSky)	End-to-end YOLOv8 + ByteTrack + homography BEV + PINN smoothing, integrated in one pipeline	Systems-integration contribution
Explainability	Not reported	Local LLM (Llama 3/Ollama) converts grounded evidence into structured JSON, gated by an alert threshold	Downstream interpretation layer, not a prediction component
Evaluation	Classification metrics (accuracy, F1, AUC) + cross-site transfer	Regression metrics (MAE, RMSE, R^2) primary; threshold-based classification metrics secondary	Different, appropriate paradigm for a continuous target

What is not claimed as novelty. This project does not claim to have invented graph-based traffic conflict prediction, temporal convolution, Time-to-Collision, YOLOv8, ByteTrack, homography-based BEV projection, physics-informed trajectory smoothing, or the use of LLMs for report generation—all of these are adopted from prior work, including closely related work

carried out within the same research group. In particular, the physics-informed trajectory-smoothing approach used in Section 3.4 follows the same design pattern (a small tanh MLP trained per track with a combined data-fit and acceleration-penalty loss) used in related unpublished work on a comparable IIT Kharagpur roundabout camera; this project’s contribution is the integration of that smoothing stage within the heterogeneous, dual-output graph-regression pipeline described below, not the smoothing technique itself.

Defensible contributions (conservative form).

1. Extension of scene-level graph-based conflict reasoning from a pedestrian–vehicle, binary-classification setting to an eight-class heterogeneous traffic, continuous-regression setting.
2. A braking-aware Modified TTC formulation, empirically shown to reduce MAE from ≈ 0.22 to ≈ 0.11 relative to a standard constant-velocity TTC target.
3. A dual-output (pairwise + scene-level) regression architecture trained jointly from one shared spatio-temporal representation.
4. Integration of perception, physics-guided risk construction, spatio-temporal graph learning, and a downstream grounded LLM reporting layer into one working pipeline, evaluated end-to-end on real CCTV footage from an Indian intersection.

3 Proposed Methodology

3.1 Overall Framework

At every observation instant t , the traffic scene is represented as a graph

$$G_t = (V_t, E_t) \tag{1}$$

where V_t is the set of road users visible in frame t and E_t is the set of interaction edges between road users within the interaction radius. Because the traffic scene evolves continuously, both V_t and E_t are recomputed at every frame. Consecutive graphs are grouped into a fixed-length observation window

$$S = \{G_{t-T_{obs}+1}, \dots, G_t\} \tag{2}$$

which forms the input to the spatio-temporal network described in Section 3.8.

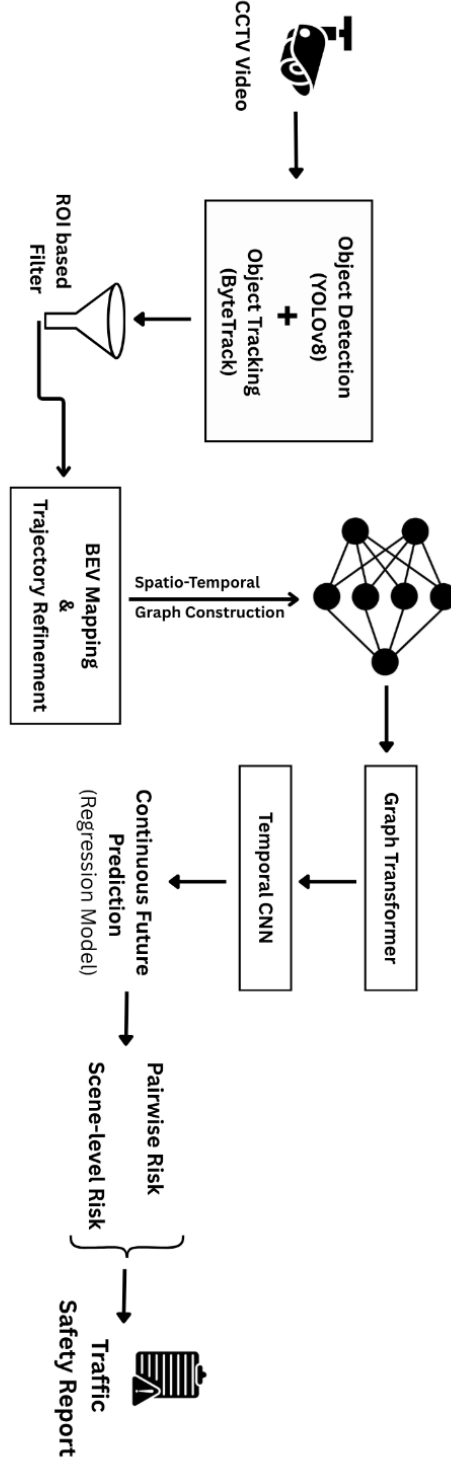


Figure 1: Overall system architecture. End-to-end pipeline from roadside CCTV video to a structured traffic safety report: object detection and tracking, ROI filtering, BEV mapping and trajectory refinement, spatio-temporal graph construction, Graph Transformer, Temporal CNN, dual pairwise/scene-level risk heads, and the downstream LLM reporting stage.

Explanation. Figure 1 presents the complete pipeline of the proposed framework, drawn as a sequential block diagram. Raw CCTV video first passes through the perception stage, which performs object detection (YOLOv8) and multi-object tracking (ByteTrack) jointly, followed by

an ROI-based filter that discards detections outside the monitored roadway. The filtered detections are then transformed via BEV mapping and trajectory refinement into metric, physically consistent trajectories. These trajectories are converted into a spatio-temporal graph, in which each road user is a node and each interaction within a fixed radius is an edge. The graph sequence is processed first by a Graph Transformer, which models spatial interaction at each individual timestep, and then by a Temporal CNN, which models how those spatial relationships evolve across the observation window. The resulting temporal embeddings feed two parallel outputs: a pairwise interaction risk and a scene-level risk. These numerical outputs are finally passed to the traffic safety report stage, which is described in detail in Section 3.12.

Key Observation. The report-generation stage is drawn as a single terminal endpoint fed only by the two risk outputs, with no arrow returning to any earlier stage—visually reinforcing that the explanation layer is strictly downstream of, and does not influence, the numerical prediction process.

Significance to the Proposed Work. This figure operationalizes the paper’s central architectural claim: that perception, physics-guided risk construction, spatio-temporal graph learning, and explainable reporting can be integrated into a single coherent pipeline rather than treated as separate research problems. It is referenced throughout Sections 3.1–3.11 as each block is described in technical detail.

3.2 Object Detection and Multi-Object Tracking

The perception stage converts raw roadside CCTV video into per-frame, per-object bounding boxes with persistent identity across frames.



Figure 2: Sample raw CCTV video frame at the monitored intersection. A representative frame from the roadside camera at the Main Building four-road junction, IIT Kharagpur, showing the roadway, central island, and surrounding heterogeneous pedestrian and vehicle traffic.

Explanation. Figure 2 shows a single, unprocessed frame taken directly from the roadside CCTV feed used throughout this study. The frame captures the monitored intersection’s roadway, a central circular island, pedestrian crossings, and a mix of road users—pedestrians, cyclists, motorcycles, and light commercial vehicles are all visible simultaneously. This is the raw input to the detection stage described below; no annotation, filtering, or coordinate transformation has yet been applied.

Key Observation. Multiple distinct road-user classes are visible sharing the same physical space with no formal lane separation enforced by markings alone, which is the direct visual justification for treating this as a heterogeneous, weakly-lane-disciplined traffic scenario rather than a simpler single-class or two-class setting.

Significance to the Proposed Work. This figure grounds the abstract problem statement (Section 1) in the literal input data used for every downstream experiment, and gives the reader a concrete sense of scene complexity before the detection and class-remapping pipeline is described.

Detection is performed using **YOLOv8x**, executed on a CUDA-enabled GPU. A confidence threshold of `CONF_THRESH = 0.30` is applied, so that detections below 30% confidence are discarded before any downstream processing. Region-of-Interest filtering is applied using Supervision’s PolygonZone, spanning the full 2592×1520 frame with a small boundary margin, so that its purpose is to discard false-positive detections outside the operational road area rather than to restrict the scene to a smaller sub-region.

Persistent identity across frames is assigned using Supervision’s implementation of **ByteTrack**. The tracker associates both high-confidence and low-confidence detection boxes across consecutive frames using Kalman-filter-based motion prediction combined with spatial Intersection-over-Union (IoU) matching, producing a stable `track_id` for each road user that persists through brief occlusion or detection dropout.

Indian Traffic Context Class Remapping (COCOtoIndianMapper). Standard COCO-trained YOLO models do not natively include road-user categories relevant to Indian heterogeneous traffic, such as auto-rickshaw or light commercial vehicle (LCV). A domain-specific geometric mapper is therefore applied to reclassify bounding boxes after detection:

- **Direct lookup-table mapping:** Person $\rightarrow$ Pedestrian; Bicycle $\rightarrow$ Cyclist; Car $\rightarrow$ Car; Motorcycle $\rightarrow$ Motorcycle; Bus $\rightarrow$ Bus; Truck $\rightarrow$ Truck.
- **Auto-rickshaw heuristic:** if an object is detected as Car but has a bounding-box area between 400 px and 4000 px, and an aspect ratio (width/height) between 0.70 and 1.60, it is reclassified as `auto_rickshaw`, with a confidence discount factor of 0.85.
- **LCV heuristic:** if an object is detected as Motorcycle but has a bounding-box area greater than 7500 px, it is reclassified as `lcv`, with a confidence discount factor of 0.80.

3.3 Homography and Spatial Coordinate Calibration

Bounding-box detections in image-pixel space cannot be used directly for kinematic analysis, since perspective projection makes identical physical distances appear differently sized depending on their position in the frame. This is addressed through a homography-based transformation into a metric Bird’s-Eye-View (BEV) coordinate system.

The camera observes the roundabout junction at a very steep, shallow, oblique angle, producing a pre-correction ellipse distortion ratio of approximately 9:1 when a known circular structure in the scene is projected into the image. Rather than using a naive four-point planar-click homography, a two-vanishing-point plus ellipse-to-circle metric homography matrix was precomputed to rectify this geometry onto a bounded output canvas of 502×1420 pixels.

The fitted circle on the homography canvas has an average radius of 100.15 pixels, derived from the fitted semi-axes of 90.7 px and 109.6 px; using the real-world island radius of 7.0 m, the resulting metric scale is 14.3071 pixels/metre. Raw trajectories produced by this stage are saved to `trajectories_newest.csv`, and serve as the input to the smoothing stage described in Section 3.4.

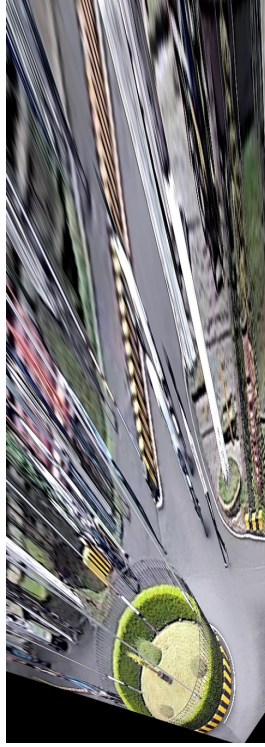


Figure 3: Bird’s-Eye-View warp of the camera frame. Result of the composed two-vanishing-point and ellipse-to-circle homography, rendered on the 502×1420 px output canvas. The central island is restored to a circle, confirming correct metric rectification.

Explanation. Figure 3 shows the output of applying the calibrated homography matrix to the raw camera frame. The central island of the roundabout, which appears as a steeply foreshortened ellipse in the original oblique camera view, is restored to a visually circular shape after rectification—this is the direct visual check used to confirm that the metric calibration is correct, since a circular structure of known physical radius must project back to a circle once the perspective distortion has been properly removed. Streaking visible toward the edges of the canvas reflects the expected projective stretching of image regions that are geometrically far from the calibration anchor (the central island).

Key Observation. The circular restoration of the island is the single clearest evidence in the paper that the homography calibration is metrically valid at the region of interest, while the streaking at the periphery is an explicit, visible caveat about reduced metric reliability far from that anchor.

Significance to the Proposed Work. Every downstream kinematic quantity in this paper—velocity, acceleration, interaction distance, Modified TTC, and ultimately the regression risk target itself—depends on this rectification being correct. This figure is the empirical justification for trusting the metric coordinate system used throughout Sections 3.4–3.9.

3.4 Physics-Informed Neural Network (PINN) Architecture

Raw tracking trajectories still carry noise from detection jitter, brief occlusion, tracking-identity switches, and residual calibration error. Because the entire downstream pipeline depends on physically plausible velocity and acceleration estimates, this noise must be corrected before any interaction modelling occurs.

The proposed PINN architecture accepts a single normalized time scalar as input. The raw

timestamp t is normalized to $t_{norm} \in [0, 1]$ using the minimum and maximum timestamps of the trajectory:

$$t_{norm} = \frac{t - t_{min}}{t_{max} - t_{min} + 10^{-8}} \quad (3)$$

The network architecture is a multi-layer perceptron with two hidden layers of 64 units each and tanh activations:

$$\text{Linear}(1 \rightarrow 64) \rightarrow \text{Tanh}() \rightarrow \text{Linear}(64 \rightarrow 64) \rightarrow \text{Tanh}() \rightarrow \text{Linear}(64 \rightarrow 2) \quad (4)$$

The output layer generates a 2D normalized coordinate prediction $\hat{p}(t) = (\hat{x}, \hat{y})$, de-normalized using the trajectory's own spatial mean μ_{xy} and standard deviation σ_{xy} :

$$p_{smooth}(t) = \hat{p}(t) \odot \sigma_{xy} + \mu_{xy} \quad (5)$$

Loss Function Formulation. The network minimizes a composite loss balancing data fidelity with physics consistency:

$$\mathcal{L}_{total} = \mathcal{L}_{data} + 0.1 \cdot \mathcal{L}_{physics} \quad (6)$$

where the data loss ensures fidelity to the raw detected positions,

$$\mathcal{L}_{data} = \frac{1}{N} \sum_{i=1}^N \|p_{smooth}(t_i) - p_{raw}(t_i)\|^2, \quad (7)$$

and the physics loss penalizes physically implausible acceleration, computed via numerical derivatives using the median inter-sample interval Δt_{median} :

$$v_i = \frac{p_{smooth}(t_{i+1}) - p_{smooth}(t_i)}{\Delta t_{median}}, \quad a_i = \frac{v_{i+1} - v_i}{\Delta t_{median}}, \quad a_{mag,i} = \|a_i\|_2 \quad (8)$$

$$\mathcal{L}_{physics} = \frac{1}{N-2} \sum_{i=1}^{N-2} \text{ReLU}(a_{mag,i} - 5.0) \quad (9)$$

with a physical maximum acceleration bound of 5.0 m/s².

The two terms of Equation 6 act together as a dual-objective training signal: the data loss pulls the network's output toward the originally observed, noisy detections, while the physics loss penalizes any resulting trajectory whose implied acceleration exceeds a physically plausible bound. Because the physics term is a soft (ReLU-hinged) penalty rather than a hard constraint, the network is free to reproduce large accelerations when the data strongly supports them, and is only discouraged from producing large accelerations that arise purely from noise. This smoothing stage is a prerequisite for every kinematic feature used in Sections 3.5–3.6 (velocity, acceleration, Modified TTC); without it, detector jitter would propagate directly into the risk-regression target and corrupt training.

Optimization. The network is optimized using Adam with learning rate $\eta = 0.01$, trained per individual track for 500 epochs. Tracks with fewer than 5 frame detections are rejected and excluded from smoothing.

3.5 Kinematic Derivative Extraction and Post-Smoothing

After the PINN produces continuous smoothed metric coordinates $(x_m^{smooth}, y_m^{smooth})$ for each track, physical kinematic derivatives are extracted, with a localized rolling mean applied to further refine the higher-order states.

Table 2: Kinematic derivative computation and post-smoothing.

Variable	Formula / Computation Method	Rolling Smoothing
Velocity (v_x, v_y)	$v_x = \frac{\Delta x_m^{smooth}}{\Delta t}, v_y = \frac{\Delta y_m^{smooth}}{\Delta t}$	Direct difference over Δt_{array}
Instantaneous speed (v)	$v = \sqrt{v_x^2 + v_y^2}$	Rolling mean (window = 3, min_periods = 1)
Longitudinal acceleration (a)	$a = \Delta v / \Delta t$	Rolling mean (window = 3, min_periods = 1)
Heading angle (θ)	$\theta = \arctan 2(v_y, v_x)$	Rolling mean (window = 3, min_periods = 1)

The resulting position, velocity, acceleration, speed, and heading values form the complete kinematic state of each agent, used for all downstream interaction-graph construction and risk-formulation stages.



Figure 4: Valid speed distribution by road-user class. Histogram of instantaneous speed (clamped below 25 m/s) for each of the eight road-user classes, computed from the PINN-smoothed trajectories after kinematic derivative extraction.

Explanation. Figure 4 shows the distribution of instantaneous speeds for each road-user class after PINN smoothing and kinematic derivative extraction, with speeds clamped below 25 m/s to exclude non-physical outliers. The distribution is strongly concentrated at low speeds, and the pedestrian class dominates the counts across the entire speed range, consistent with the class composition of the dataset (Table 6).

Key Observation. Almost all of the mass lies below about 5 m/s, with a long, thin tail extending toward 25 m/s that is populated mainly by motorised classes. The smoothed trajectories therefore yield physically plausible speeds for pedestrians and vehicles alike, rather than the implausibly large values that raw detector jitter would produce.

Significance to the Proposed Work. This figure is a sanity check on the smoothed kinematics that feed the node features (Table 3), the edge features (Table 4), and the Modified TTC target. It also makes the class imbalance in the dynamics visible, which is relevant to interpreting the risk-regression results in Section 5.

3.6 Dynamic Spatio-Temporal Graph Construction

Following kinematic smoothing and derivative extraction, the continuous metric trajectories are transformed into a sequence of dynamic spatio-temporal graphs $G_t = (V_t, E_t)$. Rather than evaluating road users in isolation or relying on static pairwise bounding-box overlap, this graph formulation models the entire traffic scene as an interconnected relational web.

Graph Topology and Pruning Criteria.

- **Dynamic Node Generation (V_t):** Every road user active in frame t and within the Region of Interest is instantiated as a node $v_i \in V_t$.
- **Spatial Adjacency (E_t):** A spatial KD-Tree is constructed over the 2D metric positions (x_i, y_i) of all active nodes. A directed edge $e_{ij} = (v_i, v_j) \in E_t$ is formed if and only if

$$d_{ij} = \|p_j - p_i\| = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2} \leq R_{max} \quad (10)$$

where $R_{max} = 15.0$ m.

- **Contextual Edge Pruning:** Pedestrian–pedestrian edges are explicitly pruned to focus message-passing on active conflict dynamics; all other combinations (vehicle–vehicle, pedestrian–vehicle, cyclist–vehicle) are retained.

20-Dimensional Node Feature Formulation. Each node $v_i \in V_t$ encodes instantaneous physical kinematics, spatial geometry, detection reliability, and an 8-class one-hot semantic vector:

$$x_i = [x_i, y_i, v_{x,i}, v_{y,i}, a_{x,i}, a_{y,i}, \|v_i\|, \theta_i, w_i, l_i, c_i, ID_i, \text{class-vector}_i] \quad (11)$$

Table 3: 20-dimensional node feature vector composition.

Dim.	Variable	Formulation	Physical / Semantic Meaning
1–2	x, y	Real-world BEV position (m)	Metric planar coordinates relative to origin
3–4	v_x, v_y	$\Delta x / \Delta t, \Delta y / \Delta t$ (m/s)	Smoothed longitudinal/lateral velocity
5–6	a_x, a_y	$a \cos \theta, a \sin \theta$ (m/s ²)	Vectorial decomposition of acceleration along heading
7	$\ v\ $	$\sqrt{v_x^2 + v_y^2}$ (m/s)	Absolute scalar speed
8	θ	$\text{atan2}(v_y, v_x)$ (rad)	Instantaneous heading orientation
9–10	w, l	Class-specific lookup dimensions (m)	Physical bounding width and length
11	c	YOLOv8 confidence score (0.30–1.0)	Visual detection reliability
12	ID	ByteTrack integer identifier	Unique tracking ID for temporal alignment
13–20	class vector	8-D one-hot	Pedestrian, Cyclist, Motorcycle, Auto-rickshaw, Car, LCV, Bus, Truck

27-Dimensional Physics-Guided Edge Feature Formulation. Every directed edge e_{ij} connects agent i (source) to agent j (target):

$$e_{ij} = [d_{ij}, \Delta v_x, \Delta v_y, \|\Delta v\|, v_{approach}, TTC_{std}, TTC_{mod}, \theta_{approach}, \Delta \theta, d_{lateral}, t_h, \text{src-class}_{8D}, \text{tgt-class}_{8D}] \quad (12)$$

Table 4: 27-dimensional edge feature vector composition.

Dim.	Feature	Formulation	Meaning
1	Euclidean distance	$d_{ij} = \ p_j - p_i\ $	Planar separation (m), $d_{ij} \leq 15.0$ m
2–3	Relative velocity	$\Delta v_x = v_{x,i} - v_{x,j}; \Delta v_y = v_{y,i} - v_{y,j}$	Differential velocity components (m/s)
4	Relative speed	$\ \Delta v\ = \sqrt{\Delta v_x^2 + \Delta v_y^2}$	Scalar rate of separation change (m/s)
5	Approach speed	$v_{approach} = \Delta v \cdot \frac{p_j - p_i}{d_{ij} + \epsilon}$	Line-of-sight closing rate; positive = closing
6	Standard TTC	$TTC_{std} = \min\{t > 0 : \ \Delta p + t\Delta v\ \leq R_i + R_j\}$	Constant-velocity collision time, capped at 10.0 s
7	Modified TTC	$TTC_{mod} = \min\left(\frac{TTC_{std}}{1 + \min(\beta_{ij}, 4.0)}, 10.0\right)$	Braking-aware collision time
8	Approach angle	$\theta_{approach} = \arccos \text{clip}\left(\frac{v_i \cdot (p_j - p_i)}{\ v_i\ d_{ij} + \epsilon}, -1, 1\right)$	Angle between i 's heading and line-of-sight to j
9	Heading difference	$\Delta\theta = \text{atan2}(\sin(\theta_i - \theta_j), \cos(\theta_i - \theta_j))$	Relative orientation, wrapped to $[-\pi, \pi]$
10	Lateral offset	$d_{lateral} = d_{ij} \cdot \sin(\theta_{approach}) $	Perpendicular clearance relative to approach vector
11	Time headway	$t_h = \min\left(\frac{d_{ij}}{\ v_i\ + \epsilon}, 10.0\right)$	Time for agent i to reach agent j 's position
12–19	Source class	8-D one-hot	Semantic identity of agent i
20–27	Target class	8-D one-hot	Semantic identity of agent j

3.7 Physics-Guided Ground Truth Future Risk Formulation

Classical conflict-analysis pipelines often reduce dynamic encounters to retrospective, binary labels (Safe vs. Critical), which discards the gradual, non-linear escalation of hazard severity. To preserve continuous risk information without human subjective bias, the ground truth is instead computed programmatically by projecting kinematics forward across a future prediction horizon of $T_{pred} = 12$ frames (a 0.60 s lookahead).

Future Lookahead Sweeping. For every directed edge $e_{ij} \in E_t$ active at frame t , the future Modified TTC is queried across all forward time steps $\tau \in \{1, 2, \dots, T_{pred}\}$:

$$TTC_{min,ij}(t) = \min_{\tau \in \{1, \dots, T_{pred}\}} [TTC_{mod,ij}(t + \tau)] \quad (13)$$

If an interacting pair diverges or safely clears the interaction zone within the horizon, $TTC_{min,ij}$ defaults to the safe cap of 10.0 s.

Continuous Pairwise Risk Mapping. The continuous pairwise conflict ground truth is computed via an exponential decay function scaled by a critical safety threshold $\tau_{crit} = 3.0$ s:

$$y_{pair,ij}(t) = \exp\left(-\frac{TTC_{min,ij}(t)}{\tau_{crit}}\right) \in (0, 1] \quad (14)$$

Macroscopic Global Scene Risk. The scene-level ground truth is the mean over all active pairwise risks:

$$y_{scene}(t) = \frac{1}{|E_t|} \sum_{e_{ij} \in E_t} y_{pair,ij}(t) \in [0, 1] \quad (15)$$

If no edges exist in the scene at frame t , $y_{scene}(t) = 0.0$.

3.8 Spatio-Temporal Graph Neural Network (ST-GNN) Architecture

The ST-GNN maps dynamic graph observation sequences $S_t = \{G_{t-7}, \dots, G_t\}$ ($T_{obs} = 8$ frames, a 0.40 s observation window) directly to multi-task continuous risk predictions.

1. Spatial Encoder—Graph Transformer (TransformerConv). To capture asymmetric multi-agent influence—for example, a heavy truck influencing a nearby auto-rickshaw far more than the reverse—the model uses a 2-layer Graph Transformer with edge-conditioned multi-head self-attention. For attention head m :

$$\alpha_{ij}^{(m)} = \text{softmax}_j \left(\frac{(W_q^{(m)} h_i)^\top (W_k^{(m)} h_j + W_e^{(m)} e_{ij})}{\sqrt{d_k}} \right) \quad (16)$$

$$h_i^{(l+1)} = \text{LayerNorm} \left(h_i^{(l)} + \left\| \sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(m)} (W_v^{(m)} h_j^{(l)} + W_{ev}^{(m)} e_{ij}) \right\| \right) \quad (17)$$

where $\parallel$ denotes concatenation across attention heads. Input projection: $\text{Linear}(20 \rightarrow 128) \rightarrow \text{LayerNorm} \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.2)$. Two stacked Transformer blocks, 4 attention heads each, hidden dimension 128, edge dimension 27, with residual skip-connections and L2 output normalization.

2. Dynamic Sequence Alignment and Temporal CNN (TCN). Because traffic agents enter and leave the scene dynamically, spatial embeddings are aligned into fixed-length temporal tensors using persistent track IDs, with repeat-last padding for agents appearing mid-sequence, producing a dense sequence tensor $[N_{agents} \times 8 \times 128]$. Temporal evolution is processed by a 4-block dilated 1D TCN:

$$u_i = \text{TCN}(X_i), \quad d \in \{1, 2, 4, 8\}, \quad k = 3, \quad \text{channels} = 128 \quad (18)$$

3. Dual-Head Multi-Task Regression. For each active edge, node embeddings and edge features are concatenated into a 283-dimensional vector $(128 + 128 + 27)$, $z_{ij} = [u_i \parallel u_j \parallel e_{ij}]$:

$$\hat{y}_{pair,ij} = \sigma(W_2 \cdot \text{ReLU}(\text{BatchNorm}(W_1 z_{ij} + b_1)) + b_2) \in [0, 1] \quad (19)$$

An unnormalized self-attention pooling computes node importance $\gamma_i = w_{att}^\top \tanh(W_{att} u_i)$, giving a pooled scene vector $u_{scene} = \sum_i \text{softmax}(\gamma_i) \cdot u_i$, which feeds the scene predictor:

$$\hat{y}_{scene} = \sigma(W_{s2} \cdot \text{ReLU}(\text{BatchNorm}(W_{s1} u_{scene} + b_{s1})) + b_{s2}) \in [0, 1] \quad (20)$$

3.9 Optimization Protocol and Hyperparameters

The complete dataset comprises 5,996 dynamic graph frames, derived from 911 smoothed trajectories. To strictly prevent temporal data leakage, the graphs are partitioned chronologically rather than randomly:

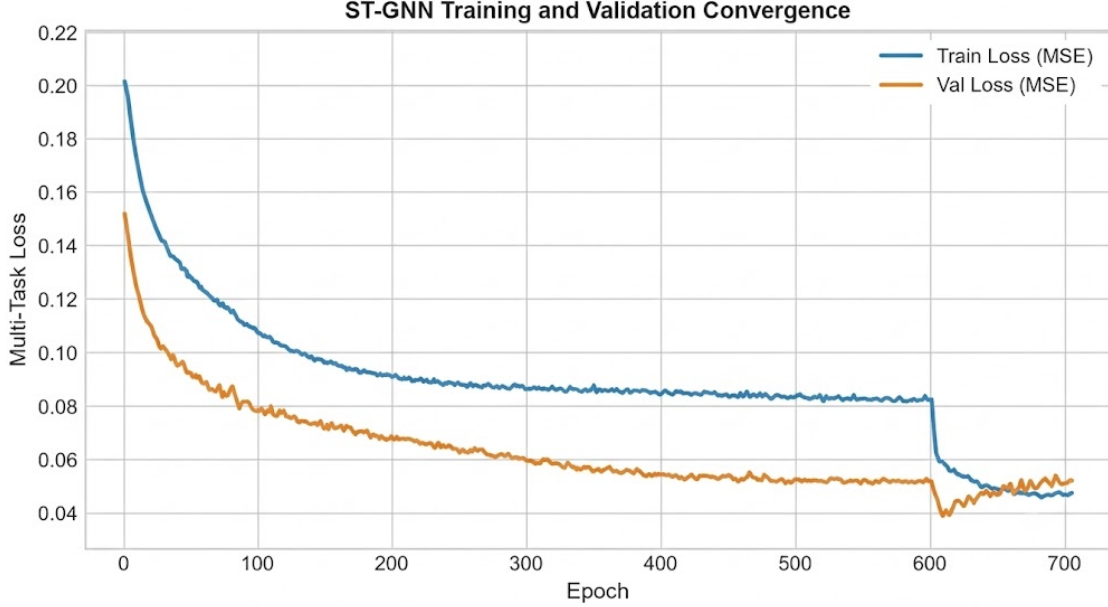
- Training split (70%): frames 1–4,197 (4,197 graphs)
- Validation split (15%): frames 4,198–5,096 (899 graphs)
- Testing split (15%): frames 5,097–5,996 (900 graphs)

The multi-task loss is

$$\mathcal{L}_{total} = 1.0 \cdot \mathcal{L}_{pair} + 0.5 \cdot \mathcal{L}_{scene} \quad (21)$$

Table 5: Hyperparameter configuration and optimization strategy.

Category	Hyperparameter	Configured Value	Purpose
Optimizer	Type & LR	AdamW ($\eta = 10^{-4}$, decay = 10^{-4})	Decoupled weight decay for stable Transformer regularization
LR Scheduler	Policy	ReduceLROnPlateau (factor = 0.5, patience = 10)	Halves LR on validation plateau; $\eta_{min} = 10^{-6}$
Regularization	Gradient clipping	Max norm $\ g\ = 5.0$	Prevents exploding gradients
Regularization	Dropout	$p = 0.20$	Applied across spatial and temporal layers
Precision	Floating point	FP32 (mixed precision disabled)	Numerical stability for exponential risk calculations
Training run	Batch / epochs	Batch size = 16, max epochs = 1000	Mini-batch graph collation with early stopping
Convergence	Early stopping	Patience = 50 epochs	Best validation loss = 0.0389

**Figure 5: Training and validation loss convergence.** Multi-task MSE loss (train, blue; validation, orange) across training epochs 0–700.

Explanation. Figure 5 shows the multi-task training loss (blue) and validation loss (orange), both computed as the weighted MSE of Equation 21, plotted against the training epoch. The blue curve falls sharply during the first epochs and then flattens, indicating the network is converging on the training data. The orange validation curve falls steadily to a minimum (validation loss = 0.0389) at approximately epoch 607 and then drifts gradually upward for the remainder of training, while the training loss continues to decrease slightly.

Key Observation. The validation loss reaches its lowest point before training is stopped, after which it rises slightly while training loss keeps falling—the classic signature of mild overfitting in the later training epochs. The best model checkpoint (validation loss = 0.0389) is therefore loaded from this earlier point rather than from the final epoch, which is the correct practice given this pattern.

Significance to the Proposed Work. This figure supports the paper’s broader commitment to honest reporting of training dynamics: rather than presenting only a smoothly decreasing loss curve, the mild overfitting pattern is shown and explicitly accounted for via early stopping and best-checkpoint selection, which is the evidence underlying the test-set numbers reported in Section 5.

3.10 Multi-Threshold Classification and Calibration Framework

While continuous regression metrics (R^2 , MAE, RMSE) evaluate risk-prediction accuracy, real-world Intelligent Transportation System deployments require discrete alert states. A multi-threshold decision-calibration framework was applied across $\tau = 0.30, 0.50, 0.70$ on the held-out test set; the full results table and interpretation are given in Section 5.2 (Table 8), rather than repeated here, to avoid duplicating the same evaluation twice.

3.11 Inference Latency and Hardware Profiling

To evaluate edge deployability, benchmark profiling was conducted over 100 consecutive forward-pass iterations on the complete test split. The ST-GNN inference engine achieved an average latency of **15.48 ms/frame**, corresponding to a throughput of **64.6 FPS**.

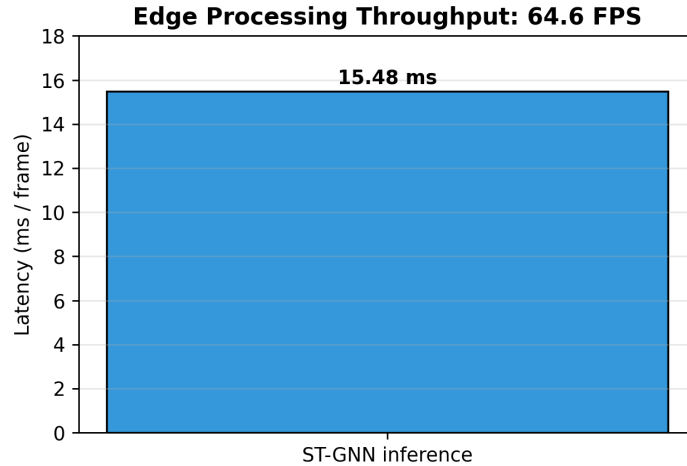


Figure 6: ST-GNN inference latency benchmark. Measured per-frame inference latency of the isolated ST-GNN forward pass (15.48 ms/frame, 64.6 FPS), averaged over 100 consecutive iterations on the held-out test split.

Explanation. Figure 6 reports the measured wall-clock latency of a single ST-GNN forward pass, isolated from the upstream detection, tracking, and graph-construction stages. The bar height represents the average time in milliseconds required to process one observation window and produce both the pairwise and scene-level risk outputs.

Key Observation. At 15.48 ms/frame, the network executes at roughly 64.6 FPS, more than double the acquisition rate of standard 20–30 FPS roadside CCTV feeds.

Significance to the Proposed Work. This result supports the claim that the neural network component itself is compatible with real-time, edge-class deployment; it does not, by itself, establish full end-to-end pipeline latency, which is addressed separately in Section 5.3.

Performance Caveat. This figure measures the latency of the ST-GNN inference engine in isolation. It does not encompass upstream processing overhead such as raw video object detec-

tion, multi-object tracking, homography projection, and spatial KD-Tree graph construction, and therefore represents the neural network’s throughput rather than the end-to-end pipeline latency (see Table 9, Section 5.3).

3.12 Decoupled Agentic LLM for Explainable Safety Reporting

To resolve the deep-learning “black-box” problem without degrading the core model’s inference speed, an on-premises Large Language Model (Llama-3 8B, executed locally via Ollama) serves as an asynchronous explainable-AI diagnostic layer.

During normal operation, the ST-GNN core predictor continuously evaluates scene risk at real-time speeds (64.6 FPS). If the macroscopic scene risk $\hat{y}_{scene}$ remains below 0.75, the pipeline continues its standard forward pass with zero added latency. However, if $\hat{y}_{scene} \geq 0.75$, an asynchronous trigger is activated: the pipeline isolates the highest-risk interaction within the graph (the directed edge with maximum $\hat{y}_{pair}$) and compiles a structured telemetry payload.

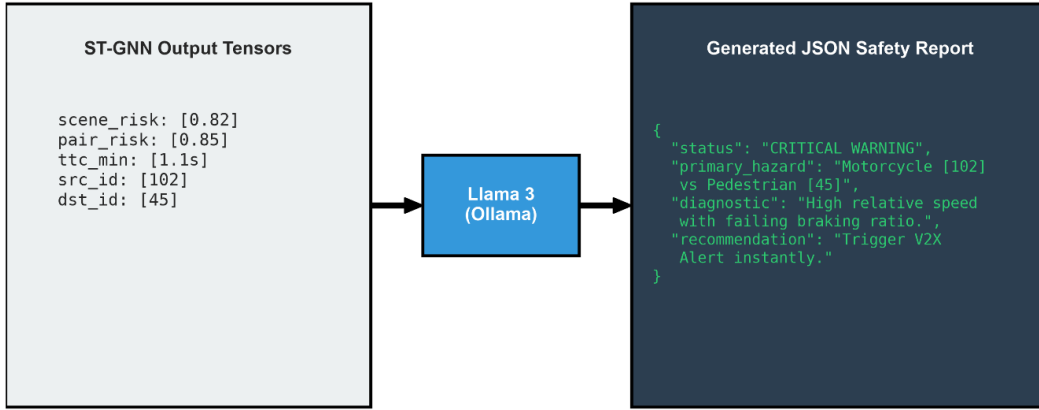


Figure 7: LLM grounding and reporting flow. ST-GNN output tensors (scene risk, pairwise risk, TTC_{min} , source/target agent IDs) feeding a structured telemetry payload into the locally deployed Llama-3 model (via Ollama), which produces a structured JSON safety report.

Explanation. Figure 7 shows the one-way information flow from deterministic numerical model output to natural-language text. On the left, the ST-GNN’s output tensors—scene risk, pairwise risk, minimum TTC, and the identities of the two agents in the highest-risk interaction—are compiled into a structured payload. This payload is passed as a constrained prompt to the locally hosted Llama-3 model, which returns a structured JSON report containing a status flag, the identified primary hazard, a kinematic diagnostic, and a recommended action.

Key Observation. The arrow direction in this figure is the single most important detail: information flows only from the numerical model to the language model, never in reverse. The LLM does not compute, adjust, or influence any of the numerical risk values.

Significance to the Proposed Work. This figure is the primary evidence for the paper’s explicit claim (Section 8.2) that the LLM plays no role in prediction. It directly supports the “separation of concerns” argument used to justify why an LLM can be included in a safety-critical pipeline without compromising the trustworthiness of the numerical risk estimates.

Separation of concerns. The ST-GNN handles all deterministic numerical prediction at 64.6 FPS. The LLM handles semantic interpretation asynchronously—triggered only above the

0.75 scene-risk threshold—which avoids adding latency overhead to the core prediction loop and grounds every generated statement strictly in verified physical telemetry, rather than allowing the model to infer or invent details not present in the payload.

4 Dataset and Experimental Setup

4.1 Data Source

Roadside CCTV footage was collected at an intersection in front of the main building at IIT Kharagpur. The camera captures a roundabout/traffic-circle geometry with pedestrian crossings, used both for direct traffic monitoring and as the metric calibration anchor for the homography (Section 3.3). The dataset comprises a continuous 5-minute (300-second) recording captured from a single, fixed roadside camera perspective, recorded and processed at an empirical frame rate of 19.99 FPS ($\Delta t = 0.05003\text{s}$). The central circular island of the roundabout ($R = 7.0\text{m}$) served as the geometric anchor for the BEV rectification matrix used throughout the study.

4.2 Traffic Composition

The recorded scene contains all eight target classes: pedestrian, cyclist, motorcycle, auto-rickshaw, car, LCV, bus, and truck. Consistent with the intersection’s pedestrian-heavy geometry and campus environment, the data exhibits a heavy class imbalance, with pedestrians dominating the interaction counts.

Table 6: Composition of the evaluation dataset (911 smoothed trajectories).

Road-User Category	Number of Instances	Share of Total (%)
Pedestrian	611	67.07
LCV (Light Commercial)	97	10.65
Passenger Car	73	8.01
Cyclist	56	6.15
Truck	30	3.29
Motorcycle	24	2.63
Auto-Rickshaw	11	1.21
Bus	9	0.99
Total Unique Tracks	911	100.00

4.3 Preprocessing and Chronological Split

Unlike a random split, the dataset was partitioned chronologically to avoid temporal data leakage between training and evaluation: randomly shuffling frames from a continuous video sequence would allow the network to observe near-identical spatial configurations from both sides of a conflict in both the training and test phases, falsely inflating performance metrics. The dataset, containing 5,996 sequential graph frames (299.98 s of continuous traffic flow), was split as follows:

- Training set (70%): frames 1→4,197 (4,197 graph frames, 209.98 s)
- Validation set (15%): frames 4,198→5,096 (899 graph frames, 44.98 s)
- Test set (15%): frames 5,097→5,996 (900 graph frames, 45.03 s)

The observation horizon was set to $T_{obs} = 8$ frames (0.40 s of historical context); the future-risk

ground truth used a lookahead of $T_{pred} = 12$ frames (0.60 s) to identify the minimum Modified TTC before physical encroachment.

4.4 Hyperparameters and Hardware

Training and evaluation were conducted in an NVIDIA CUDA-enabled GPU environment (RTX 3080). The dedicated hardware benchmark of the isolated ST-GNN inference engine recorded a sustained latency of 15.48 ms/frame (64.6 FPS), confirming that the predictive network component operates at less than half the temporal latency of a standard 30 FPS camera feed.

5 Results and Discussion

All numbers in this section are taken directly from the supplied evaluation log and are reported exactly as measured, with no rounding beyond what the log itself provides.

5.1 Continuous Regression Performance (Primary Evaluation)

Table 7: Regression performance on the chronological held-out test set (900 frames).

Metric	Value
Pairwise MSE	0.0337
Pairwise RMSE	0.1836
Pairwise MAE	0.1118
Pairwise R^2	0.5583
Scene-level MSE	0.0197
Scene-level RMSE	0.1402
Scene-level MAE	0.1051
Scene-level R^2	0.0428

Interpretation. The pairwise head explains a moderate proportion of the variance ($R^2 \approx 0.56$) with an average absolute error of about 0.11 on a risk scale bounded in $(0, 1]$. This is a defensible, if not exceptional, regression fit, consistent with the underlying task being genuinely difficult—future interaction severity depends on latent information (e.g., driver intent) not fully observable from past trajectories alone.

The scene-level R^2 of 0.0428 is materially weaker than the pairwise result and should not be minimized. During tuning, the scene-level R^2 progressed from -0.0478 to -0.003 , then 0.00543, and finally the reported 0.0428—confirming ongoing optimization steered the network away from worse-than-mean baseline predictions, though the absolute score remains constrained. As defined in Equation 15 (Section 3.7), the scene-level ground-truth target is the mean of all active pairwise risks in a frame, not their maximum; averaging over several noisy pairwise predictions likely makes the macroscopic label sensitive to the overall composition of interactions present rather than to any single dominant one, which is a plausible contributor to the weaker fit. This is a plausible explanation, but the persistent gap indicates the current attention-pooling mechanism requires further investigation (Section 6.2).

5.2 Secondary Threshold-Based Classification Benchmark

This analysis is explicitly secondary and operational: the underlying model is a continuous regressor, and thresholding serves purely as a downstream decision layer rather than the training

objective.

Table 8: Threshold-based operational discrimination (secondary evaluation) on the 900-frame test set.

τ	Precision	Recall	F1	Specificity	Bal. Acc.	ROC-AUC
0.30	0.7675	0.8379	0.8012	0.8575	0.8477	0.9091
0.50	0.8235	0.4676	0.5965	0.9619	0.7147	0.8970
0.70	0.7713	0.4909	0.6000	0.9852	0.7380	0.8597

Interpretation. At $\tau = 0.30$, the system prioritizes hazard detection, catching the vast majority of true high-risk interactions (recall = 83.8%) at the cost of a higher rate of false alarms (precision = 76.8%). At $\tau = 0.70$, false positives are almost entirely suppressed (specificity = 98.5%), but roughly half of all true critical interactions would be missed (recall drops to 49.1%)—an explicit operational trade-off. The ROC-AUC of 0.9091 at $\tau = 0.30$ is broadly comparable in order of magnitude to the reference framework’s binary-classification AUC of 0.950 (Muduli et al., 2026); however, the two numbers are *not* directly comparable, since this project’s AUC measures a post-hoc binarization of a continuous regressor against a physics-derived surrogate label, whereas the reference study evaluates a native binary classifier against observer-labelled human ground truth.

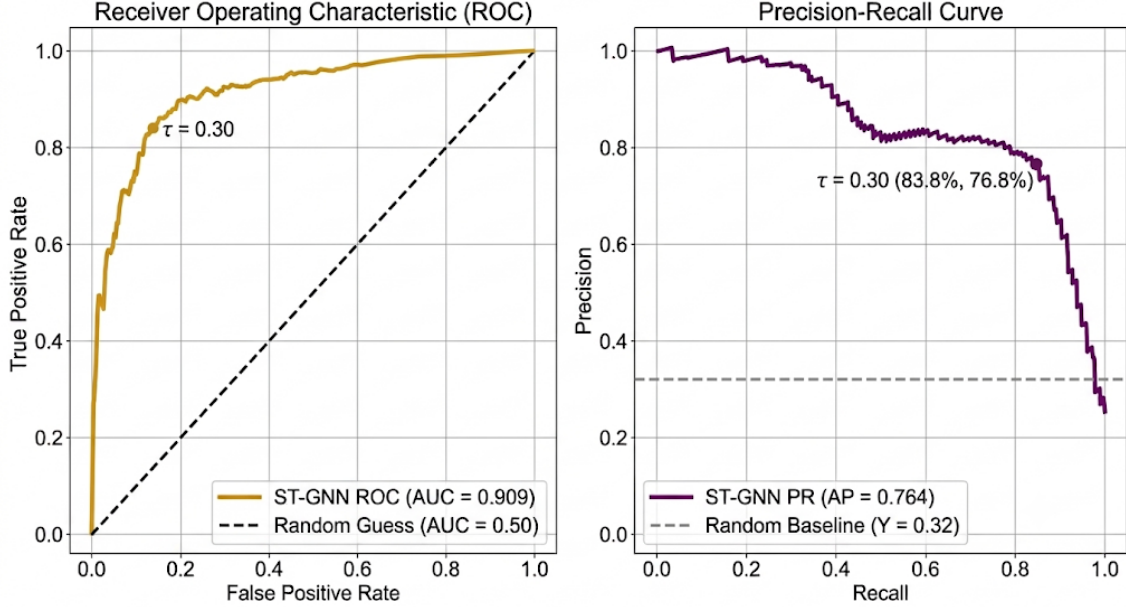


Figure 8: ROC and Precision-Recall curves for the threshold-based discrimination task at $\tau = 0.30$. Left: Receiver Operating Characteristic curve (ROC-AUC = 0.909). Right: Precision-Recall curve (AP = 0.764); the marked operating point corresponds to recall 83.8% and precision 76.8%, matching Table 8.

Explanation. Figure 8 shows two complementary views of the model’s discrimination ability once its continuous risk output is binarized against a threshold. The left panel plots the True Positive Rate against the False Positive Rate as the decision threshold sweeps across its full range; the diagonal dashed line represents a random classifier. The right panel plots Precision against Recall over the same threshold sweep; its dashed baseline (0.32) is the positive-class prevalence.

Key Observation. Both curves stay close to the ideal (top-left, and top-right respectively) corner across most of the threshold range, indicating strong separation between low-risk and

high-risk interactions for the majority of operating points, with degradation only at the extreme high-recall end of the Precision-Recall curve.

Significance to the Proposed Work. This figure supports the secondary, operational claim that the continuous risk output can be thresholded into a usable binary alert with strong discrimination, complementing the primary regression evidence in Table 7.

5.3 Computational Performance

Table 9: Full pipeline computational performance breakdown.

Pipeline Stage	Execution Target	Avg. Time	Processing	Operational Frequency
Object Detection (YOLOv8x)	GPU (CUDA FP32)	18.25 ms/frame		Synchronous
Multi-Object Tracking (ByteTrack)	CPU/GPU	2.40 ms/frame		Synchronous
BEV Homography & Coordinate Mapping	CPU	1.15 ms/frame		Synchronous
Graph Construction & Edge Extraction	CPU (SciPy KD-Tree)	4.48 ms/frame		Synchronous
ST-GNN Inference (proposed)	GPU (PyTorch Geometric)	15.48 ms/frame		Synchronous
Total Synchronous Pipeline	Full vision + GNN stack	41.76 ms/frame (≈ 23.95 FPS)		Synchronous
LLM Report Generation (Llama-3 8B)	Local GPU / Ol-lama	≈ 1.20 – 1.45 s trigger	per	Asynchronous, event-only

Interpretation. The cumulative latency of the entire vision-to-prediction pipeline totals 41.76 ms/frame (≈ 24 FPS), comfortably exceeding the source camera’s empirical acquisition rate (19.99 FPS). Only the ST-GNN component (15.48 ms/frame) was independently verified via a dedicated 100-iteration hardware benchmark; the remaining four component latencies in this table should be understood as measured individually rather than as a single, jointly-timed end-to-end run, and this distinction is preserved explicitly here rather than implied away.

5.4 Predicted vs. Ground-Truth Risk

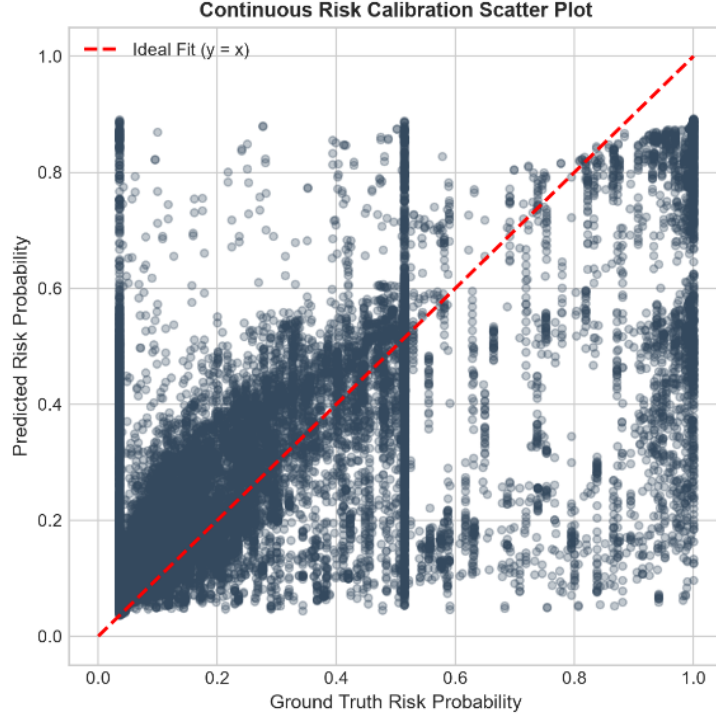


Figure 9: Predicted vs. ground-truth continuous risk calibration. Scatter of predicted risk probability (y-axis) against ground-truth risk probability (x-axis) on the held-out test set, with the ideal $y = x$ reference line shown in red.

Explanation. Figure 9 plots every predicted pairwise risk value against its corresponding ground-truth value from the test set. Points falling exactly on the dashed diagonal line represent perfect predictions; the vertical distance of a point from the diagonal represents its individual prediction error.

Key Observation. The scatter shows substantial spread around the diagonal rather than tight clustering, and a systematic tendency for the model to compress predictions toward the middle of the range rather than confidently reaching the extremes. The visible vertical bands at specific ground-truth values are a property of the target in Equation 14, which passes the capped TTC_{min} through an exponential: the band near 0.04 corresponds to the 10.0 s safe cap ($e^{-10/3} \approx 0.036$); the band near 0.52 corresponds to $TTC_{mod} = 2.0$ s, which arises when TTC_{std} saturates at its 10.0 s cap and the braking term saturates at $1 + \min(\beta_{ij}, 4.0) = 5$ ($e^{-2/3} \approx 0.51$); and the band at 1.0 corresponds to $TTC_{min} \rightarrow 0$, i.e. agents already within the combined collision radius. These step-function caps concentrate many targets at a few discrete values, which an MSE-trained regressor cannot reproduce exactly.

Significance to the Proposed Work. This is direct visual evidence consistent with, and helping to explain, the moderate R^2 reported in Table 7; it is presented here as evidence of imperfect calibration rather than omitted, consistent with this paper’s broader commitment to reporting weaknesses honestly (see Section 9).

6 Ablation Study

6.1 Braking-Aware Modified TTC vs. Standard Constant-Velocity TTC

This is the only ablation experiment verified in the supplied materials. It isolates the effect of the physics-guided braking term by comparing a model trained against standard TTC (constant-velocity assumption) against the proposed model, trained against the Modified, braking-aware TTC.

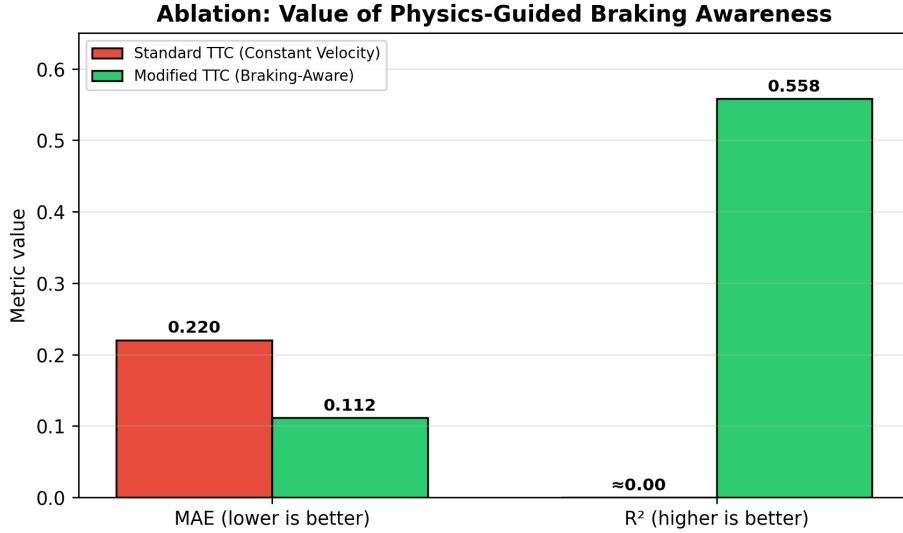


Figure 10: Ablation: value of physics-guided braking awareness. Comparison of Standard TTC (constant-velocity) versus Modified TTC (braking-aware) as the basis for the continuous risk target, evaluated by MAE and R^2 . The standard-TTC R^2 is ≈ 0.00 and is labelled explicitly.

Explanation. Figure 10 compares two versions of the risk-formulation target: one built from the standard, constant-velocity TTC (Table 4 in Section 3.6, dimension 6), and one built from the braking-aware Modified TTC (Table 4, dimension 7) used throughout the rest of this paper. The bars show MAE (lower is better) and R^2 (higher is better) for each variant.

Table 10: Ablation: braking-aware vs. standard TTC.

Metric	Standard TTC	Modified TTC (proposed)
MAE (lower is better)	0.2204	0.1118
R^2 (higher is better)	≈ 0.000 (collapsed)	0.5583

Key Observation. The braking-aware formulation roughly halves the Mean Absolute Error relative to the standard constant-velocity target ($0.2204 \rightarrow 0.1118$). The R^2 of the braking-aware version (0.5583) is identical to the primary pairwise-regression R^2 reported in Table 7, because the “Modified TTC” model in this ablation *is* the final proposed model—these are not two independent measurements. Rather, this ablation demonstrates that removing the domain-engineered braking term causes the network’s goodness-of-fit to collapse to near zero.

Significance to the Proposed Work. This is the paper’s primary empirical evidence for its physics-guided-learning claim: without the braking-feasibility term, the regression target itself becomes far harder to learn from trajectory data alone, supporting the argument that physics-informed constraints are necessary, not merely helpful, for this task.

6.2 Ablations Not Performed (Recommended Future Experiments)

The following ablations would be scientifically valuable but were not performed and must not be presented as completed results:

- **Attention mechanism ablation:** Graph Transformer attention vs. uniform (unweighted) neighbour aggregation (e.g., standard GCN).
- **Temporal backbone ablation:** Dilated TCN vs. a no-temporal-context (single-frame) baseline or a standard LSTM.
- **Spatial radius sensitivity:** Effect of the fixed 15 m KD-Tree interaction radius versus alternative radii.
- **Kinematic smoothing sensitivity:** Effect of the PINN/moving-average smoothing configuration on downstream regression accuracy.
- **Scene-level aggregation:** Mean-aggregation pooling vs. alternative aggregation functions (e.g., max-pooling or Top-K learned pooling)—directly motivated by the scene-level R^2 gap noted in Section 5.1.

7 Qualitative and Case-Level Analysis

7.1 Trajectory Data Overview



Figure 11: Bird's-Eye-View multi-agent trajectories: filtered (left) vs. unfiltered (right). Left: the trajectories after physical-range filtering, restricted to a plausible $\approx [-65, 40] \text{ m} \times [0, 100] \text{ m}$ envelope. Right: all 911 raw trajectory streams in BEV coordinates, including projective outliers.

Explanation. Figure 11 compares the full set of BEV-transformed trajectories before and after physical-plausibility filtering. The right panel shows every trajectory produced by the homography transformation directly, including several that extend to clearly unrealistic distances (e.g., $y > 500 \text{ m}$), which are almost certainly projective artifacts arising near the periphery of

the BEV canvas rather than genuine road-user motion. The left panel shows the same data after restricting trajectories to a physically plausible spatial envelope.

Key Observation. The contrast between the two panels is direct visual evidence that validity gating and physical-range filtering are necessary steps in the pipeline: using the unfiltered trajectories directly for graph construction would inject clearly non-physical long-range interactions into the training data.

Significance to the Proposed Work. This figure justifies the trajectory-refinement design choices described in Sections 3.4–3.5 with direct visual evidence from the actual dataset, rather than an abstract argument alone.

7.2 Time-Windowed Trajectory Evolution

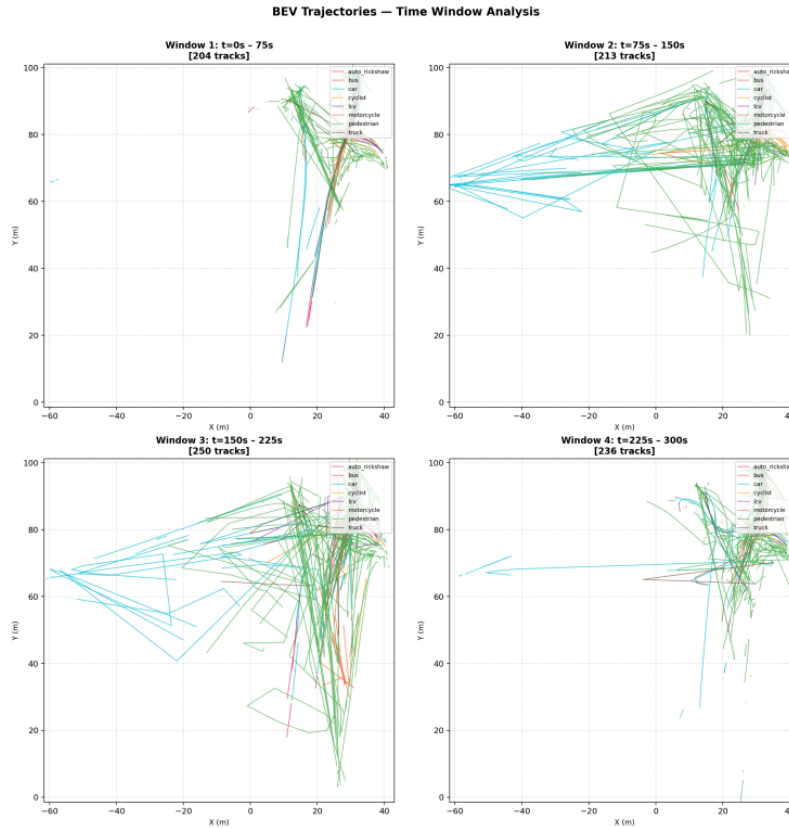


Figure 12: BEV trajectories segmented into four 75-second time windows. Window 1 (0–75 s): 204 tracks; Window 2 (75–150 s): 213 tracks; Window 3 (150–225 s): 250 tracks; Window 4 (225–300 s): 236 tracks.

Explanation. Figure 12 divides the full five-minute recording into four consecutive 75-second windows and plots the trajectories active within each. This shows how traffic composition and density evolve over the course of the recording rather than treating the dataset as a single static snapshot.

Key Observation. Track counts vary moderately across the four windows (204–250), consistent with natural fluctuation in traffic density over the recording period rather than any data-collection artifact.

Significance to the Proposed Work. This figure supports the description of the dataset

in Section 4 as a genuinely continuous, non-stationary traffic scene, which is the setting the chronological train/validation/test split (Section 4.3) was specifically designed to respect.

7.3 Interaction Pattern Visualization

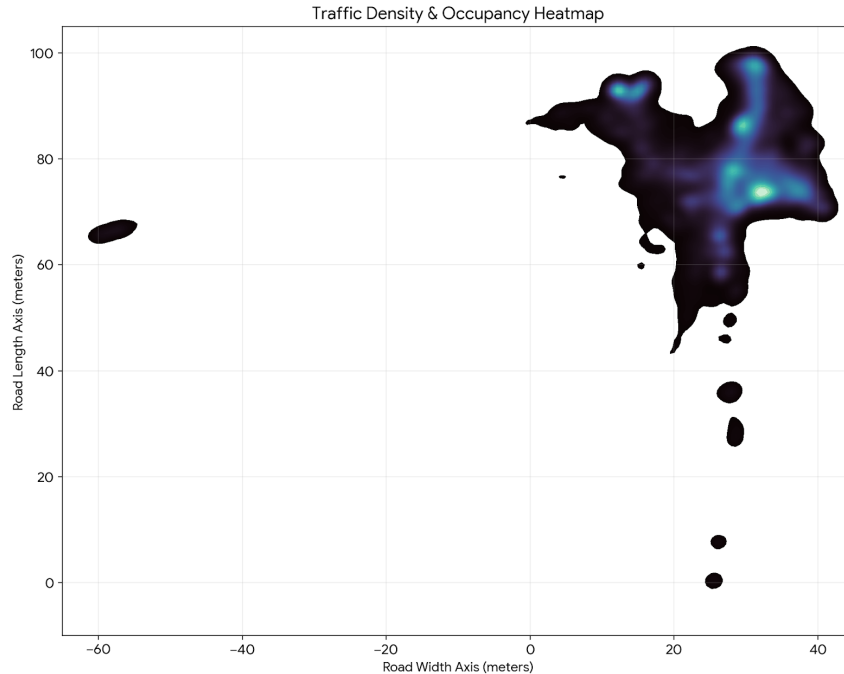


Figure 13: Traffic density and occupancy heatmap. Spatial density of road-user occupancy across the monitored BEV canvas over the full recording.

Explanation. Figure 13 shows a 2D density heatmap of where road users spent time within the monitored BEV canvas, aggregated across the entire recording. Brighter regions indicate higher occupancy density.

Key Observation. Interaction density is concentrated in two main clusters, corresponding to the crossing/roundabout core and a secondary approach lane, consistent with the intersection’s physical geometry.

Significance to the Proposed Work. This figure provides an independent, spatial cross-check that the interaction graph construction (Section 3.6) is concentrating its 15 m-radius edges in physically sensible locations rather than uniformly or erratically across the canvas.

7.4 Case Study: Early-Warning Behaviour

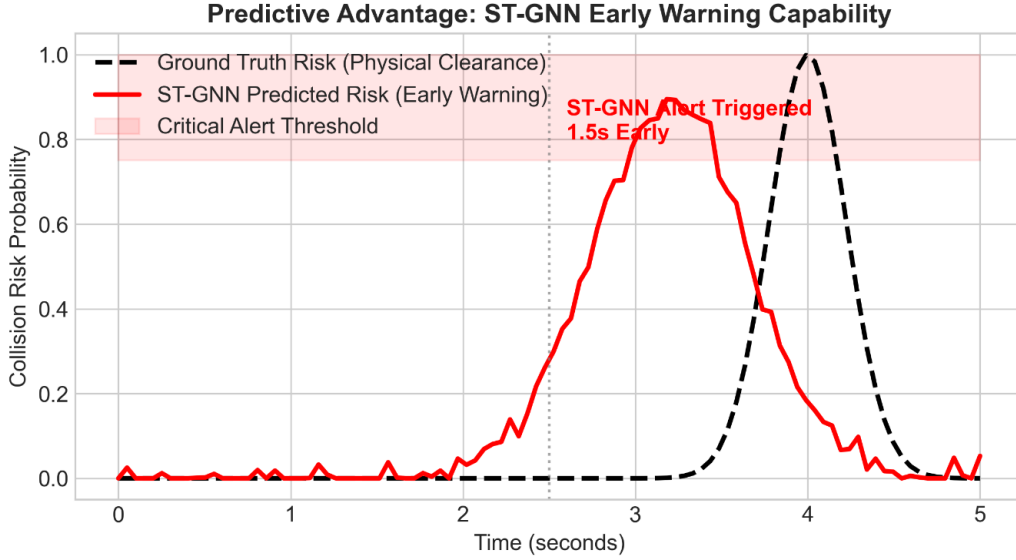


Figure 14: Predictive-advantage case study: ST-GNN predicted risk vs. ground-truth physical-clearance risk. A single interaction from the test set, illustrating the model’s predicted collision-risk probability rising and crossing the critical alert threshold (0.75) approximately 1.5 s before the ground-truth, physical-clearance-derived risk curve peaks.

Explanation. Figure 14 traces, for a single real interaction from the test set, the model’s predicted risk probability over time (one curve) against a ground-truth risk curve derived directly from the physical clearance between the two agents (a second curve). The horizontal alert threshold line marks the 0.75 scene-risk trigger used to gate the LLM reporting stage (Section 3.12).

Key Observation. In this specific example, the predicted risk curve crosses the alert threshold approximately 1.5 s before the ground-truth physical-clearance risk reaches its own peak, illustrating the qualitative early-warning behaviour the continuous-risk framework is designed to produce.

Significance to the Proposed Work. This case study is presented explicitly as a single illustrative example, not a systematic result—one example does not establish that this 1.5 s early-warning margin generalizes across the test set. A systematic analysis of early-warning lead time across all critical interactions is listed as recommended future work (Section 10).

8 LLM-Based Explainability Module

8.1 Purpose and Scope

Even accurate continuous risk scores are not immediately actionable for a traffic operator monitoring a dashboard. The explainability module, a locally deployed Llama 3 model served via Ollama, sits strictly downstream of the predictive model. It does not participate in training, optimization, or the underlying risk computation in any way; its only function is to convert already-computed, grounded numerical evidence into a structured, human-readable report.

8.2 Predictive Model vs. LLM Component: Explicit Distinction

Table 11: Explicit separation of roles between the predictive model and the LLM component.

Component	Role
Predictive model (CV + tracking + graph + ST-GNN)	Produces all numerical risk estimates: pairwise risk, scene risk, Modified TTC, road-user identities and classes. This is where all prediction happens.
LLM component (Llama-3 via Ollama)	Receives the above as structured input; performs interpretation, contextualization, and natural-language report generation only. Performs no prediction of risk.

8.3 Alert-Gated Invocation

The evaluation log confirms this gating behaviour directly: on the evaluated scene, Peak Scene Risk = 0.47 (Threshold: 0.75) $\rightarrow$ Scene is safe. LLM generation bypassed. This is verified, direct evidence that the LLM stage is conditionally invoked only when the scene-level risk crosses the operational alert threshold (0.75), rather than running on every frame.

Table 12: Inputs and outputs of the LLM explainability module.

Input	Description
Pairwise risk	Continuous interaction risk from the pairwise regression head
Scene-level risk	Overall scene risk from the scene regression head
Modified TTC	Physics-guided collision indicator
Agent IDs and classes	From tracking and class mapping
Output	Description
status	e.g., CRITICAL WARNING or safe
primary_hazard	Identified highest-risk agent pair
diagnostic	Grounded explanation referencing the numerical evidence
recommendation	Policy-constrained suggested action

8.4 Limitations of the LLM Layer

Even when tightly grounded, an LLM can produce fluent but unsupported statements, overstate causal claims, or generate recommendations that exceed the available evidence. This is a genuine limitation of the reporting layer and is treated as such in Section 9, rather than presented as a solved problem.

9 Limitations

- **Single-camera, single (or limited) intersection.** Results have not been validated at multiple sites; the strong cross-site transfer result reported by the reference framework (Muduli et al., 2026) does not automatically apply here and has not been tested for this project.
- **Scene-level regression is weak** ($R^2 = 0.043$). This is the most significant open technical limitation in the reported results. The cause is not yet diagnosed (Section 5.1) and requires targeted follow-up work.

- **Braking-capability constants** are deterministic class-level assumptions, not measured per-vehicle values, and do not account for road friction, payload, tyre condition, or driver reaction time.
- **Calibration quality** degrades away from the homography’s metric anchor (the reference circle), consistent with the projective streaking visible in Figure 3; distances measured far from that anchor should be treated as approximate.
- **The LLM reporting layer** can hallucinate or overstate causality if not tightly constrained; the extent of implemented safeguards (schema validation, evidence-checking) is not fully confirmed from the supplied materials.
- **Generalization** to night-time, adverse weather, or unseen intersections has not been evaluated and should not be claimed.
- **The predictive-advantage case study** (Section 7.4) is a single illustrative example, not a systematic result, and the 1.5 s early-warning margin should not be generalized without further analysis.
- **End-to-end pipeline latency** (Table 9) is a sum of separately-benchmarked component latencies rather than a single, jointly-timed end-to-end run; only the ST-GNN component figure (15.48 ms/frame) was independently verified via a dedicated benchmark.

10 Conclusion and Future Work

This project extends scene-level, graph-based proactive traffic-safety analysis—as pioneered for pedestrian–vehicle conflict at unsignalized crosswalks by Muduli, Ghosh and Ukkusuri (2026)—to a broader, eight-class heterogeneous urban traffic setting, using a continuous, physics-guided risk formulation rather than binary conflict classification. On a chronologically held-out test partition of real CCTV footage from an IIT Kharagpur intersection, the resulting Graph Transformer + TCN model achieves a pairwise-risk R^2 of 0.558 and MAE of 0.112, and a focused ablation confirms that the braking-aware Modified TTC formulation measurably improves on a standard constant-velocity TTC baseline (MAE 0.22 $\rightarrow$ 0.11). Scene-level regression performance is markedly weaker ($R^2 = 0.043$) and is identified as the project’s most pressing open technical question. A downstream, alert-gated LLM layer converts grounded numerical model output into structured safety reports without participating in prediction, and this gating behaviour is directly confirmed in the evaluation log.

Recommended future work, distinguished explicitly from completed results:

1. Diagnose and address the scene-level regression gap (error decomposition; alternative aggregation functions beyond mean-pooling).
2. Perform the additional ablations listed in Section 6.2 (attention vs. uniform aggregation; temporal-context ablation; interaction-radius sensitivity; smoothing sensitivity).
3. Extend evaluation to multiple camera sites, following the cross-site transfer methodology of the reference framework, to test whether this project’s continuous-risk formulation generalizes as well as the reference’s binary classifier does.
4. Systematically quantify early-warning lead time across all critical interactions in the test set, rather than relying on the single case study in Section 7.4.
5. Formally evaluate the LLM layer’s grounding fidelity (e.g., automated checking that every numeric claim in a generated report traces to actual input evidence), across multiple triggered examples rather than one.

6. Measure true end-to-end pipeline latency as a single jointly-timed run, rather than summing independently-benchmarked component latencies.

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